

Mode 10 — Square

C-DP8 (C8×C8) → DP16 (8×8) → DP16 (8×4) → DP8 (8×4) → DP8 (4×4) → add-and-square

Step 1 — C-DP8 (C8×C8)

$$\text{C-DP8 (C8} \times \text{C8)} = D = \sum_{i=0}^7 a_i b_i, \quad a_i = a_i^{re} + j a_i^{im}, \quad b_i = b_i^{re} + j b_i^{im}$$

Step 2 — DP16 (8×8)

$$\text{Re}(D) = \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{re} - \sum_{i=0}^7 a_i^{im} b_i^{im}}_{\text{DP16 (8} \times \text{8)}}$$

$$\text{Im}(D) = \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{im} + \sum_{i=0}^7 a_i^{im} b_i^{re}}_{\text{DP16 (8} \times \text{8)}}$$

Step 3 — DP16 (8×4)

$$\text{Re}(D) = 2^4 \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{re,hi} - \sum_{i=0}^7 a_i^{im} b_i^{im,hi}}_{\text{DP16 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{re,lo} - \sum_{i=0}^7 a_i^{im} b_i^{im,lo}}_{\text{DP16 (8} \times \text{4)}}$$

$$\text{Im}(D) = 2^4 \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{im,hi} + \sum_{i=0}^7 a_i^{im} b_i^{re,hi}}_{\text{DP16 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{im,lo} + \sum_{i=0}^7 a_i^{im} b_i^{re,lo}}_{\text{DP16 (8} \times \text{4)}}$$

Step 4 — DP8 (8×4)

$$\text{Re}(D) = 2^4 \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{re,hi}}_{\text{DP8 (8} \times \text{4)}} - 2^4 \underbrace{\sum_{i=0}^7 a_i^{im} b_i^{im,hi}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{re,lo}}_{\text{DP8 (8} \times \text{4)}} - \underbrace{\sum_{i=0}^7 a_i^{im} b_i^{im,lo}}_{\text{DP8 (8} \times \text{4)}}$$

$$\text{Im}(D) = 2^4 \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{im,hi}}_{\text{DP8 (8} \times \text{4)}} + 2^4 \underbrace{\sum_{i=0}^7 a_i^{im} b_i^{re,hi}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{im,lo}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i^{im} b_i^{re,lo}}_{\text{DP8 (8} \times \text{4)}}$$

Step 5 — DP8 (4×4)

$$\begin{aligned} \text{Re}(D) = & 2^8 \underbrace{\sum_{i=0}^7 a_{H,i}^{re} b_i^{re,hi}}_{\text{DP8 (4} \times \text{4)}} + 2^4 \underbrace{\sum_{i=0}^7 a_{L,i}^{re} b_i^{re,hi}}_{\text{DP8 (4} \times \text{4)}} - 2^8 \underbrace{\sum_{i=0}^7 a_{H,i}^{im} b_i^{im,hi}}_{\text{DP8 (4} \times \text{4)}} - 2^4 \underbrace{\sum_{i=0}^7 a_{L,i}^{im} b_i^{im,hi}}_{\text{DP8 (4} \times \text{4)}} \\ & + 2^4 \underbrace{\sum_{i=0}^7 a_{H,i}^{re} b_i^{re,lo}}_{\text{DP8 (4} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_{L,i}^{re} b_i^{re,lo}}_{\text{DP8 (4} \times \text{4)}} - 2^4 \underbrace{\sum_{i=0}^7 a_{H,i}^{im} b_i^{im,lo}}_{\text{DP8 (4} \times \text{4)}} - \underbrace{\sum_{i=0}^7 a_{L,i}^{im} b_i^{im,lo}}_{\text{DP8 (4} \times \text{4)}} \end{aligned}$$

$$\text{Im}(D) = 2^8 \underbrace{\sum_{i=0}^7 a_{H,i}^{re} b_i^{im,hi}}_{\text{DP8 (4} \times \text{4)}} + 2^4 \underbrace{\sum_{i=0}^7 a_{L,i}^{re} b_i^{im,hi}}_{\text{DP8 (4} \times \text{4)}} + 2^8 \underbrace{\sum_{i=0}^7 a_{H,i}^{im} b_i^{re,hi}}_{\text{DP8 (4} \times \text{4)}} + 2^4 \underbrace{\sum_{i=0}^7 a_{L,i}^{im} b_i^{re,hi}}_{\text{DP8 (4} \times \text{4)}}$$

$$+ 2^4 \underbrace{\sum_{i=0}^7 a_{H,i}^{re} b_i^{im,lo}}_{\text{DP8 (4x4)}} + \underbrace{\sum_{i=0}^7 a_{L,i}^{re} b_i^{im,lo}}_{\text{DP8 (4x4)}} + 2^4 \underbrace{\sum_{i=0}^7 a_{H,i}^{im} b_i^{re,lo}}_{\text{DP8 (4x4)}} + \underbrace{\sum_{i=0}^7 a_{L,i}^{im} b_i^{re,lo}}_{\text{DP8 (4x4)}}$$

Step 6 — Add-and-square (substitute each DP8 (4x4) = $\frac{1}{2} [\sum (x_i + b_i)^2 - \sum x_i^2 - \sum b_i^2]$)

$$\begin{aligned} \text{Re}(D) = & 2^8 \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 (a_{H,i}^{re} + b_i^{re,hi})^2 - \sum_{i=0}^7 (a_{H,i}^{re})^2 - \sum_{i=0}^7 (b_i^{re,hi})^2 \right)}_{\text{DP8 (4x4)}} \\ & + 2^4 \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 (a_{L,i}^{re} + b_i^{re,hi})^2 - \sum_{i=0}^7 (a_{L,i}^{re})^2 - \sum_{i=0}^7 (b_i^{re,hi})^2 \right)}_{\text{DP8 (4x4)}} \\ & + 2^8 \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 (a_{H,i}^{im} - b_i^{im,hi})^2 - \sum_{i=0}^7 (a_{H,i}^{im})^2 - \sum_{i=0}^7 (b_i^{im,hi})^2 \right)}_{\text{DP8 (4x4)}} \\ & + 2^4 \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 (a_{L,i}^{im} - b_i^{im,hi})^2 - \sum_{i=0}^7 (a_{L,i}^{im})^2 - \sum_{i=0}^7 (b_i^{im,hi})^2 \right)}_{\text{DP8 (4x4)}} \\ & + 2^4 \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 (a_{H,i}^{re} + b_i^{re,lo})^2 - \sum_{i=0}^7 (a_{H,i}^{re})^2 - \sum_{i=0}^7 (b_i^{re,lo})^2 \right)}_{\text{DP8 (4x4)}} \\ & + \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 (a_{L,i}^{re} + b_i^{re,lo})^2 - \sum_{i=0}^7 (a_{L,i}^{re})^2 - \sum_{i=0}^7 (b_i^{re,lo})^2 \right)}_{\text{DP8 (4x4)}} \\ & + 2^4 \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 (a_{H,i}^{im} - b_i^{im,lo})^2 - \sum_{i=0}^7 (a_{H,i}^{im})^2 - \sum_{i=0}^7 (b_i^{im,lo})^2 \right)}_{\text{DP8 (4x4)}} \\ & + \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 (a_{L,i}^{im} - b_i^{im,lo})^2 - \sum_{i=0}^7 (a_{L,i}^{im})^2 - \sum_{i=0}^7 (b_i^{im,lo})^2 \right)}_{\text{DP8 (4x4)}} \end{aligned}$$

$$\begin{aligned} \text{Im}(D) = & 2^8 \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 (a_{H,i}^{re} + b_i^{im,hi})^2 - \sum_{i=0}^7 (a_{H,i}^{re})^2 - \sum_{i=0}^7 (b_i^{im,hi})^2 \right)}_{\text{DP8 (4x4)}} \\ & + 2^4 \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 (a_{L,i}^{re} + b_i^{im,hi})^2 - \sum_{i=0}^7 (a_{L,i}^{re})^2 - \sum_{i=0}^7 (b_i^{im,hi})^2 \right)}_{\text{DP8 (4x4)}} \\ & + 2^8 \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 (a_{H,i}^{im} + b_i^{re,hi})^2 - \sum_{i=0}^7 (a_{H,i}^{im})^2 - \sum_{i=0}^7 (b_i^{re,hi})^2 \right)}_{\text{DP8 (4x4)}} \\ & + 2^4 \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 (a_{L,i}^{im} + b_i^{re,hi})^2 - \sum_{i=0}^7 (a_{L,i}^{im})^2 - \sum_{i=0}^7 (b_i^{re,hi})^2 \right)}_{\text{DP8 (4x4)}} \end{aligned}$$

$$\begin{aligned}
& + 2^4 \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 (a_{H,i}^{re} + b_i^{im,lo})^2 - \sum_{i=0}^7 (a_{H,i}^{re})^2 - \sum_{i=0}^7 (b_i^{im,lo})^2 \right)}_{\text{DP8 (4x4)}} \\
& + \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{L,i}^{re} + b_i^{im,lo})^2 - \sum_{i=0}^7 (a_{L,i}^{re})^2 - \sum_{i=0}^7 (b_i^{im,lo})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^4 \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 (a_{H,i}^{im} + b_i^{re,lo})^2 - \sum_{i=0}^7 (a_{H,i}^{im})^2 - \sum_{i=0}^7 (b_i^{re,lo})^2 \right)}_{\text{DP8 (4x4)}} \\
& + \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{L,i}^{im} + b_i^{re,lo})^2 - \sum_{i=0}^7 (a_{L,i}^{im})^2 - \sum_{i=0}^7 (b_i^{re,lo})^2 \right)}_{\text{DP8 (4x4)}}
\end{aligned}$$

Step 7 — All components

Re(D):

$$\begin{aligned}
\text{PE} &= 2^8 \sum_{i=0}^7 (a_{H,i}^{re} + b_i^{re,hi})^2 + 2^4 \sum_{i=0}^7 (a_{L,i}^{re} + b_i^{re,hi})^2 + 2^8 \sum_{i=0}^7 (a_{H,i}^{im} - b_i^{im,hi})^2 + 2^4 \sum_{i=0}^7 (a_{L,i}^{im} - b_i^{im,hi})^2 \\
&+ 2^4 \sum_{i=0}^7 (a_{H,i}^{re} + b_i^{re,lo})^2 + \sum_{i=0}^7 (a_{L,i}^{re} + b_i^{re,lo})^2 + 2^4 \sum_{i=0}^7 (a_{H,i}^{im} - b_i^{im,lo})^2 + \sum_{i=0}^7 (a_{L,i}^{im} - b_i^{im,lo})^2 \\
\alpha &= 2^8 \sum_{i=0}^7 (a_{H,i}^{re})^2 + 2^4 \sum_{i=0}^7 (a_{L,i}^{re})^2 + 2^8 \sum_{i=0}^7 (a_{H,i}^{im})^2 + 2^4 \sum_{i=0}^7 (a_{L,i}^{im})^2 \\
&+ 2^4 \sum_{i=0}^7 (a_{H,i}^{re})^2 + \sum_{i=0}^7 (a_{L,i}^{re})^2 + 2^4 \sum_{i=0}^7 (a_{H,i}^{im})^2 + \sum_{i=0}^7 (a_{L,i}^{im})^2 \\
\beta &= 2^8 \sum_{i=0}^7 (b_i^{re,hi})^2 + 2^4 \sum_{i=0}^7 (b_i^{re,hi})^2 + 2^8 \sum_{i=0}^7 (b_i^{im,hi})^2 + 2^4 \sum_{i=0}^7 (b_i^{im,hi})^2 \\
&+ 2^4 \sum_{i=0}^7 (b_i^{re,lo})^2 + \sum_{i=0}^7 (b_i^{re,lo})^2 + 2^4 \sum_{i=0}^7 (b_i^{im,lo})^2 + \sum_{i=0}^7 (b_i^{im,lo})^2
\end{aligned}$$

Im(D):

$$\begin{aligned}
\text{PE} &= 2^8 \sum_{i=0}^7 (a_{H,i}^{re} + b_i^{im,hi})^2 + 2^4 \sum_{i=0}^7 (a_{L,i}^{re} + b_i^{im,hi})^2 + 2^8 \sum_{i=0}^7 (a_{H,i}^{im} + b_i^{re,hi})^2 + 2^4 \sum_{i=0}^7 (a_{L,i}^{im} + b_i^{re,hi})^2 \\
&+ 2^4 \sum_{i=0}^7 (a_{H,i}^{re} + b_i^{im,lo})^2 + \sum_{i=0}^7 (a_{L,i}^{re} + b_i^{im,lo})^2 + 2^4 \sum_{i=0}^7 (a_{H,i}^{im} + b_i^{re,lo})^2 + \sum_{i=0}^7 (a_{L,i}^{im} + b_i^{re,lo})^2 \\
\alpha &= 2^8 \sum_{i=0}^7 (a_{H,i}^{re})^2 + 2^4 \sum_{i=0}^7 (a_{L,i}^{re})^2 + 2^8 \sum_{i=0}^7 (a_{H,i}^{im})^2 + 2^4 \sum_{i=0}^7 (a_{L,i}^{im})^2 \\
&+ 2^4 \sum_{i=0}^7 (a_{H,i}^{re})^2 + \sum_{i=0}^7 (a_{L,i}^{re})^2 + 2^4 \sum_{i=0}^7 (a_{H,i}^{im})^2 + \sum_{i=0}^7 (a_{L,i}^{im})^2 \\
\beta &= 2^8 \sum_{i=0}^7 (b_i^{im,hi})^2 + 2^4 \sum_{i=0}^7 (b_i^{im,hi})^2 + 2^8 \sum_{i=0}^7 (b_i^{re,hi})^2 + 2^4 \sum_{i=0}^7 (b_i^{re,hi})^2 \\
&+ 2^4 \sum_{i=0}^7 (b_i^{im,lo})^2 + \sum_{i=0}^7 (b_i^{im,lo})^2 + 2^4 \sum_{i=0}^7 (b_i^{re,lo})^2 + \sum_{i=0}^7 (b_i^{re,lo})^2
\end{aligned}$$